COM115 Week 6

Matrices

Refer to the Python code examples in the lecture notes, and write code to answer the following questions.

Enter your code for each Exercise in the cell beneath it. You can add additional code cells if you prefer.

In [ ]:
```
#run this cell to import numpy
import numpy as np
```

Exercise 1

Let

$$A = \begin{bmatrix} 1 & 4 \\ 3 & -1 \\ 0 & 7 \end{bmatrix}, B = \begin{bmatrix} 2 & 1 & 3 \\ -5 & 6 & 1 \end{bmatrix}, C = \begin{bmatrix} -3 \\ 9 \\ 0 \\ 9 \end{bmatrix}, D = \begin{bmatrix} 2 & 1 \\ -3 & 0 \end{bmatrix}$$

(a) Represent each matrix in Python and determine its order.

(b) Display the transpose of each matrix.

(c) What is $A + B^T$?

(d) What is $D^T - 2D$?

In [ ]:
```
#Enter your code here for Exercise 1 here
```

Exercise 2

If possible carry out the following matrix multiplications based on the matrices from Exercise 1.

(a) BD

(b) $D^T A^T$

(c) CC^T

(d) D^2, D^3, D^4, D^5

(e) Show that $AB \neq BA$

(f) Show that $(AB)^T = B^T A^T$

Let

$$E = \begin{bmatrix} 2 & 2 \\ 1 & 5 \\ 4 & 3 \end{bmatrix}$$

(g) Show that $A(BE) = (AB)E$

In [ ]:

Exercise 3

Show how you would create the following matrices using special functions rather than having to list all the values in your code.

(a) $G = \begin{bmatrix} 7 & 7 & 7 & 7 & 7 \\ 7 & 7 & 7 & 7 & 7 \\ 7 & 7 & 7 & 7 & 7 \\ 7 & 7 & 7 & 7 & 7 \\ 7 & 7 & 7 & 7 & 7 \end{bmatrix}$

In [ ]:

(b) $H = \begin{bmatrix} 2 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \\ -1 & 2 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \\ -1 & -1 & 2 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \\ -1 & -1 & -1 & 2 & -1 & -1 & -1 & -1 & -1 & -1 \\ -1 & -1 & -1 & -1 & 2 & -1 & -1 & -1 & -1 & -1 \\ -1 & -1 & -1 & -1 & -1 & 2 & -1 & -1 & -1 & -1 \\ -1 & -1 & -1 & -1 & -1 & -1 & 2 & -1 & -1 & -1 \\ -1 & -1 & -1 & -1 & -1 & -1 & -1 & 2 & -1 & -1 \\ -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & 2 & -1 \\ -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & 2 \end{bmatrix}$

In [ ]:

(c) $K = \begin{bmatrix} 2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$

In [ ]:

Exercise 4

(a) Find the determinant of each of the matrices in Exercise 3.

(b) If possible, find the inverse of the matrices G and H in Exercise 3.

In [ ]:

Exercise 5

Solve the following set of linear equations for x, y and z:

$$x + y + z = 3 \tag{1}$$
$$2x + 7y + 2z = 15 \tag{2}$$
$$-3x + 4y + z = 0 \tag{3}$$

In [ ]: